



Solar-Powered Remote Water Level Relay

PROJECT DOCUMENT

Generated by: W Si, forlook88@gmail.com

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PROJECT DETAILS

Project name:

Solar-Powered Remote Water Level Relay

Project description:

This project implements a solar-powered water level sensor system designed to monitor reservoir or tank levels and wirelessly control a remote water pump located 2km away. The sensor detects when the water reaches the halfway mark and transmits a signal to the pump station. The pump is programmed to automatically shut off if it does not receive a signal from the sensor for 10 seconds, ensuring reliable and autonomous operation.

The system utilizes a small solar panel to power the water level sensor, enabling off-grid deployment. Wireless communication is established between the sensor and the pump controller to cover the 2km distance. The project focuses on efficient power management and robust signal transmission for remote water management applications.

Application(s):

IoT, Agriculture, Industrial Automation

Selected CAD tool:

Altium Designer

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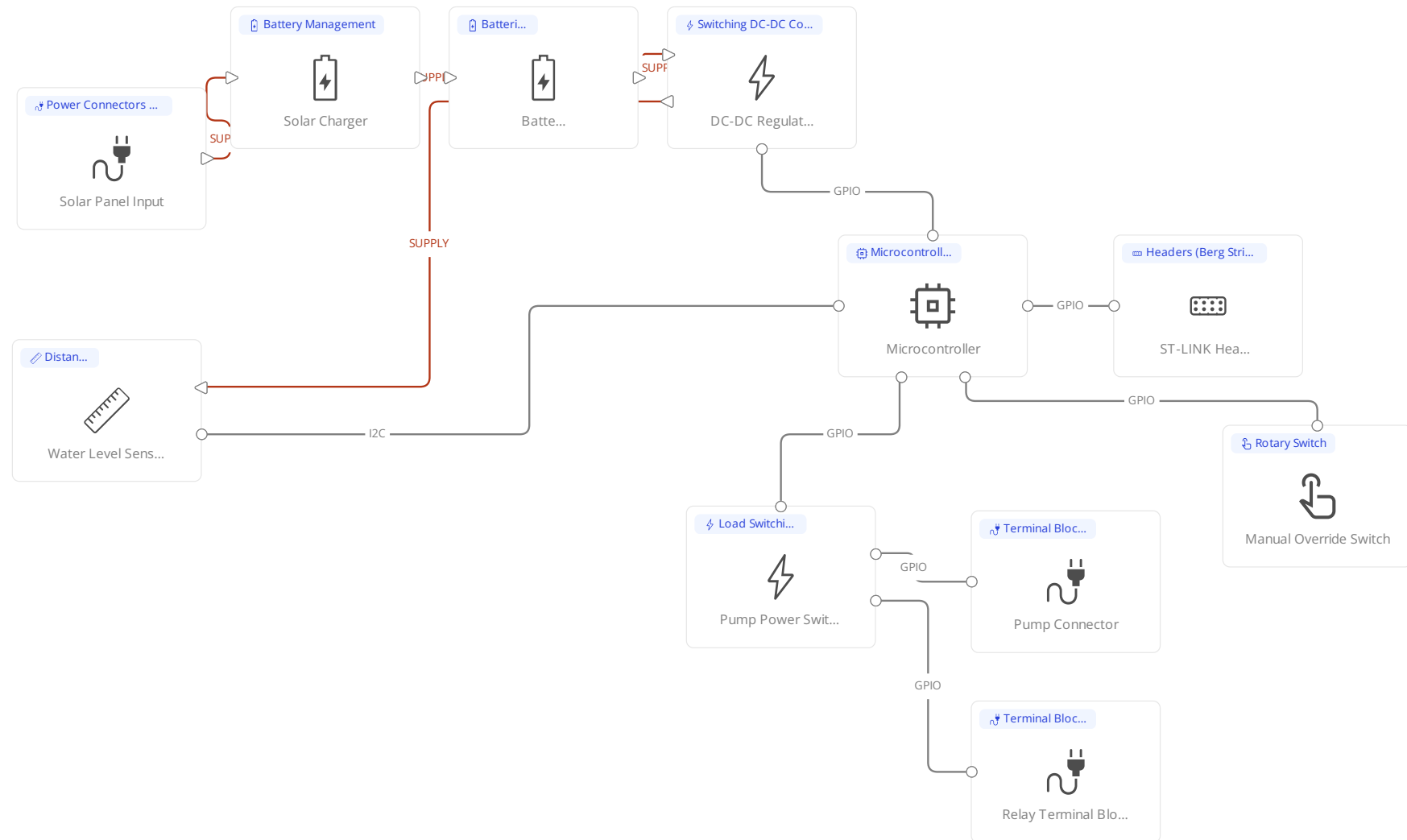
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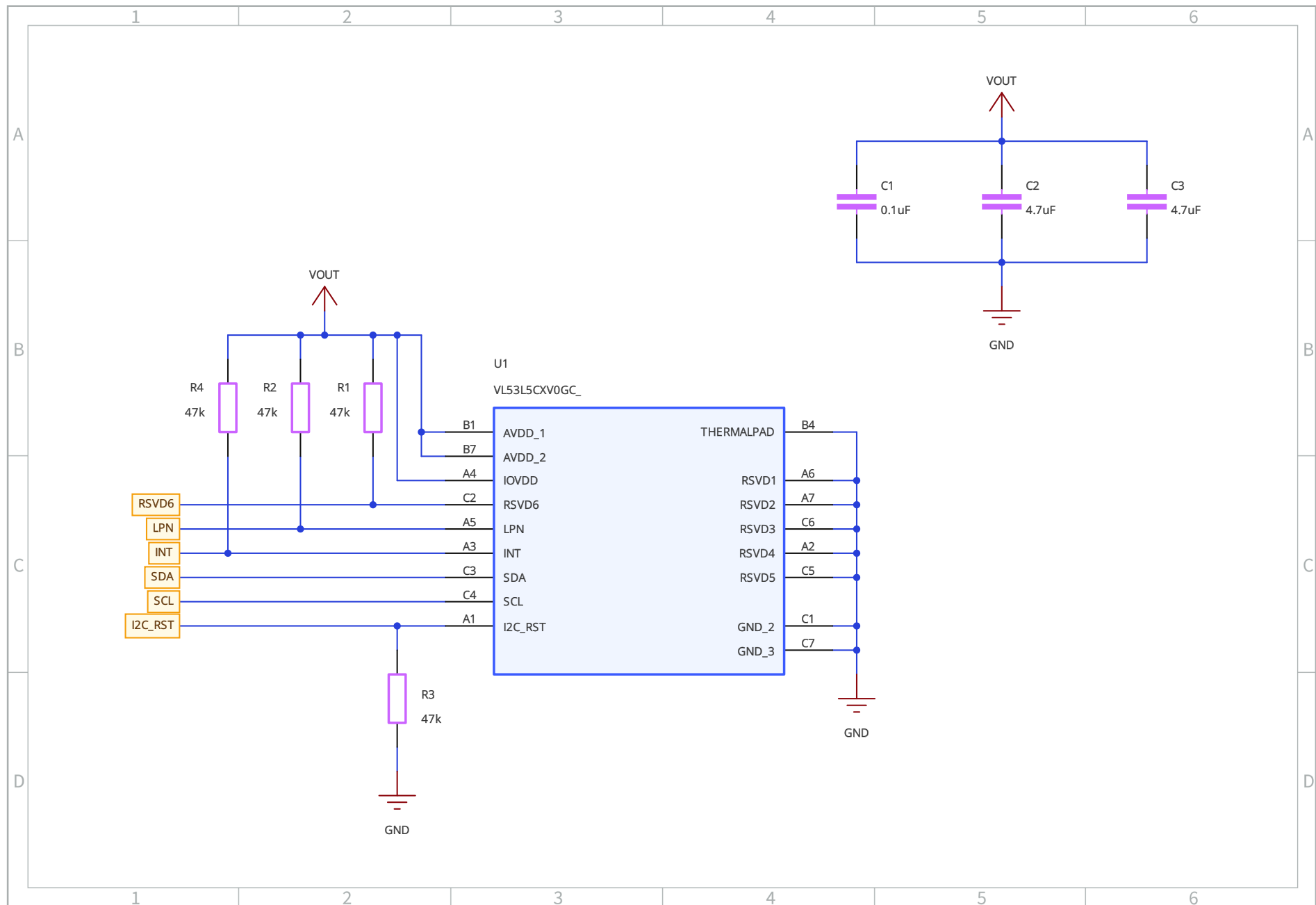
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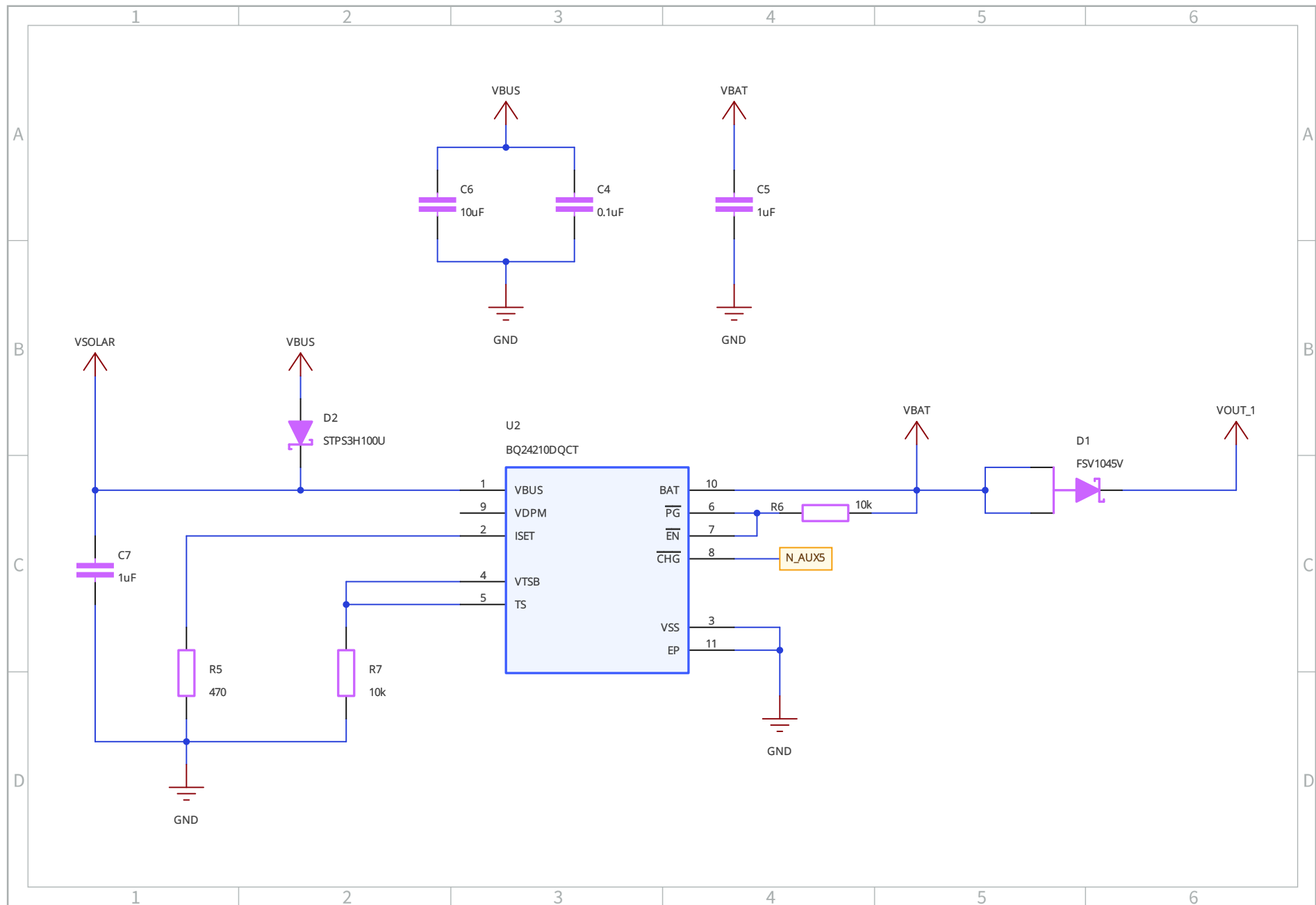
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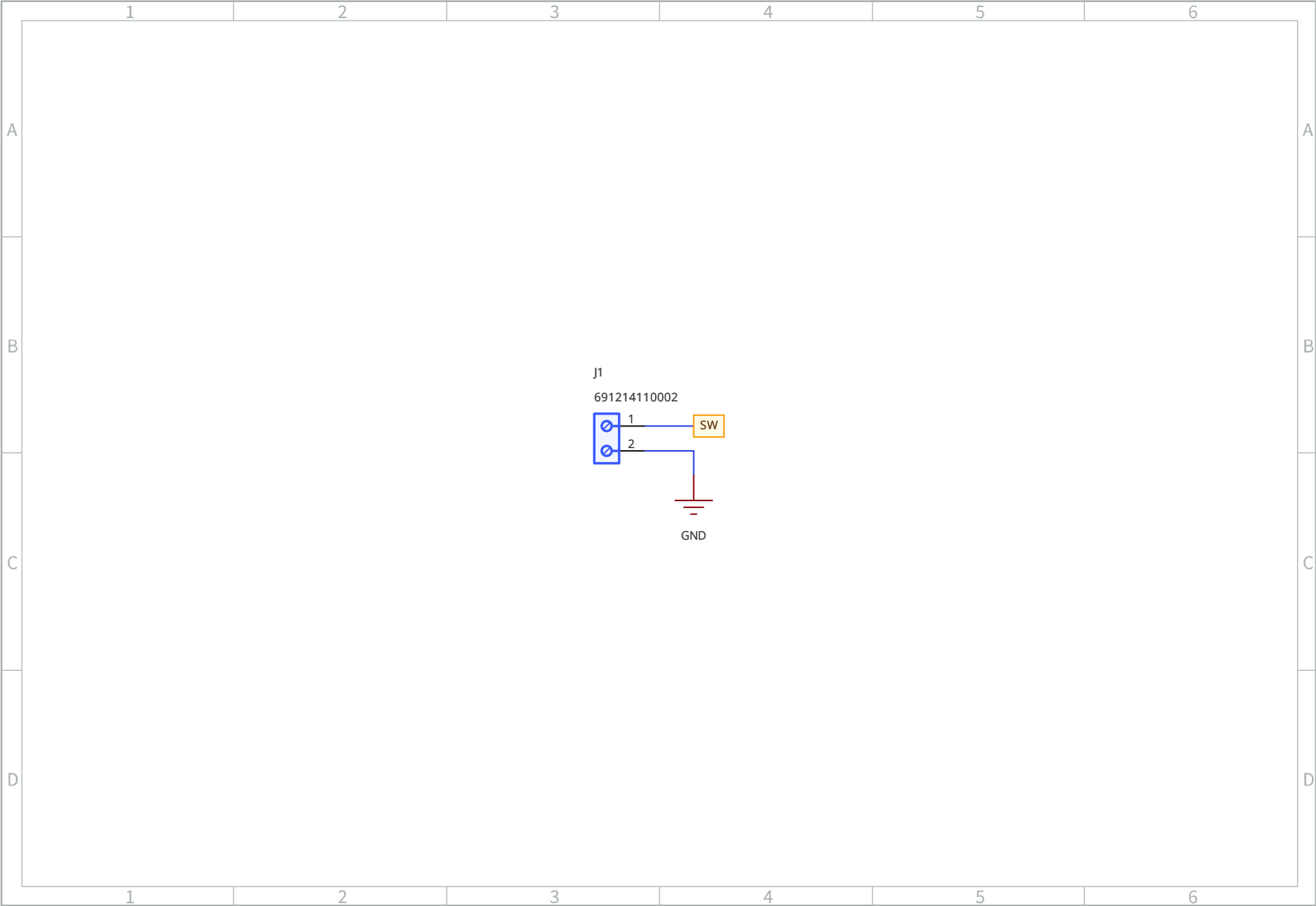
DISCLAIMER

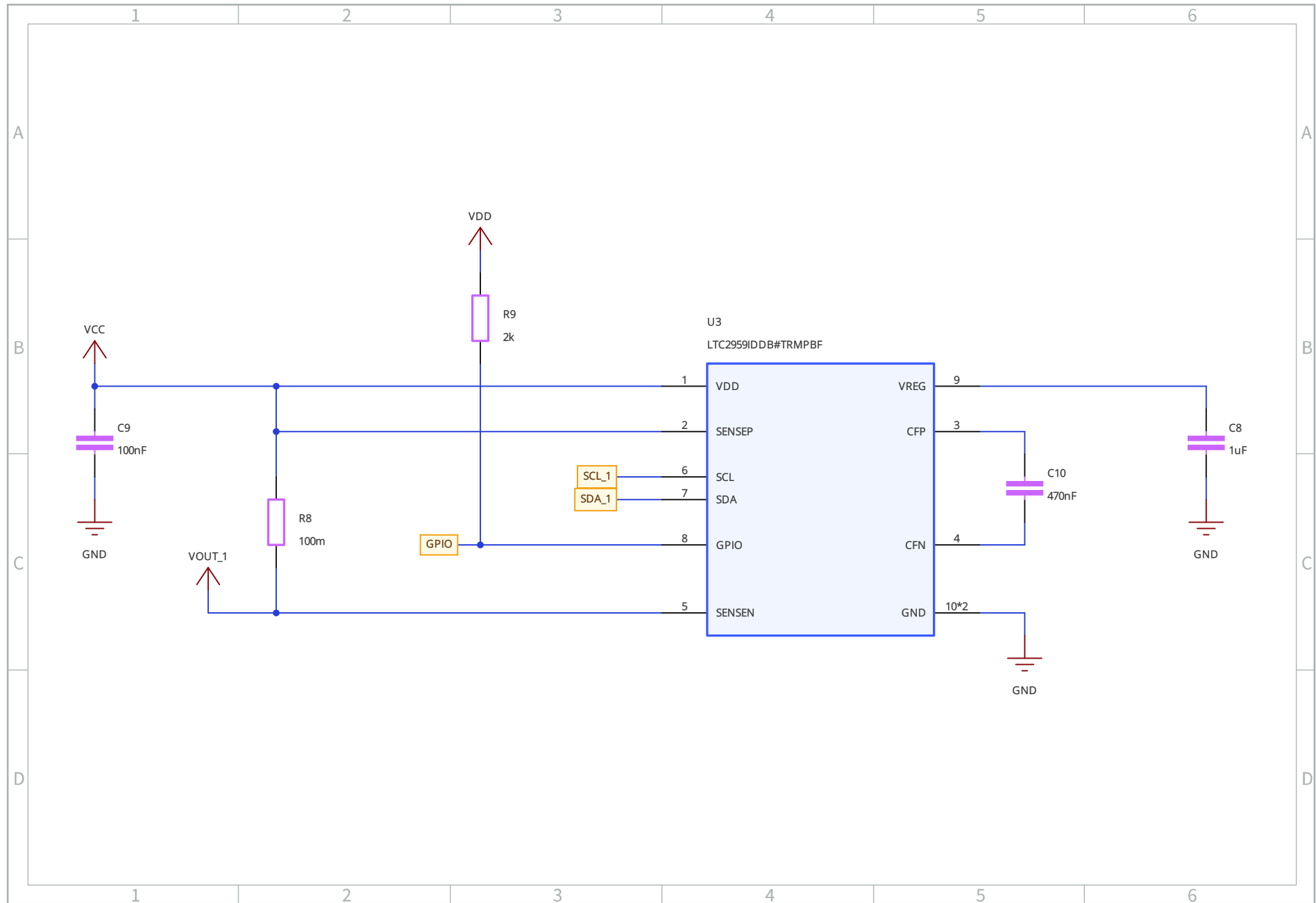
Project Title, Project Description, and Block Diagram may be AI-Generated. Schematics, BOM, and Components are not AI-Generated.

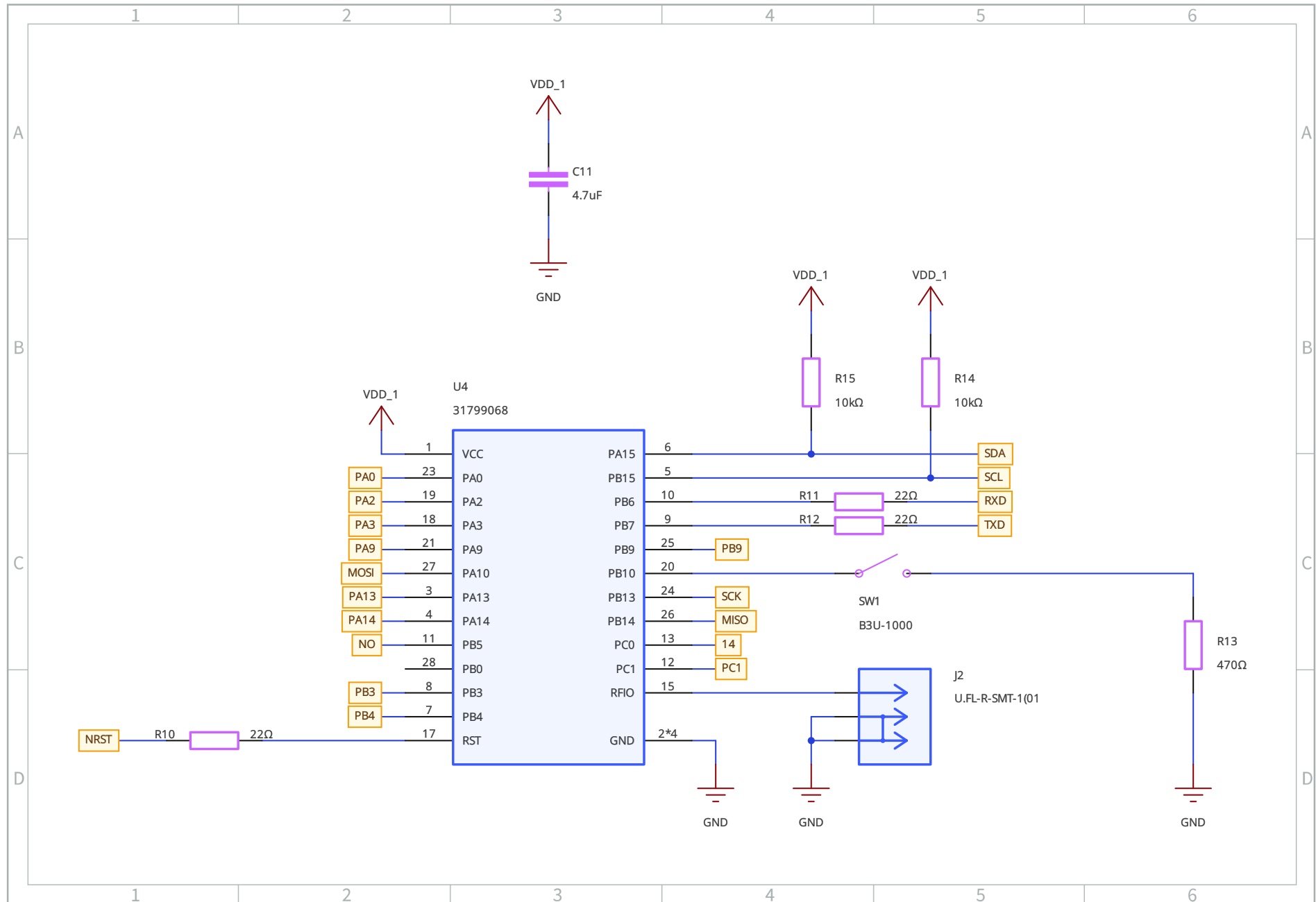


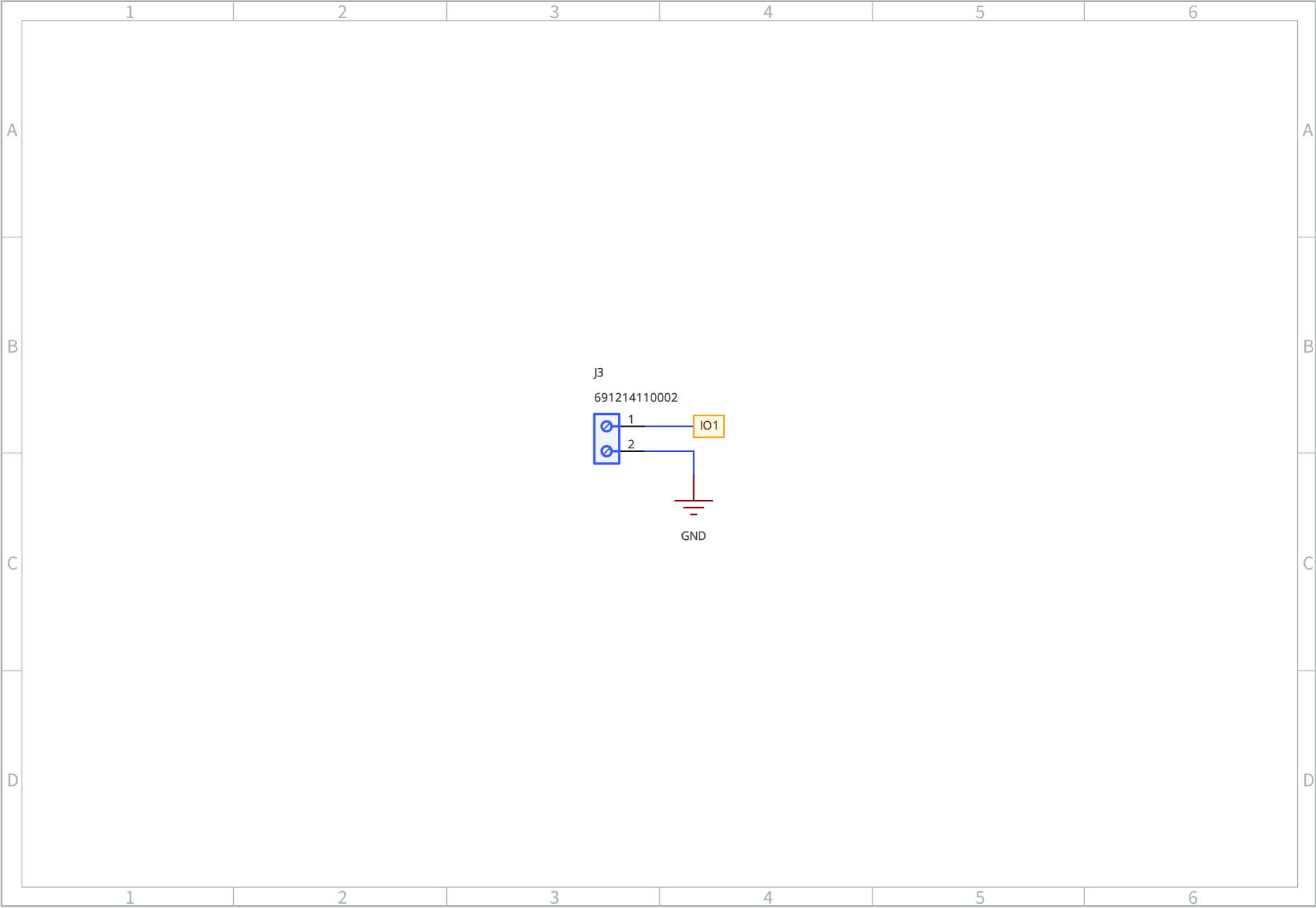


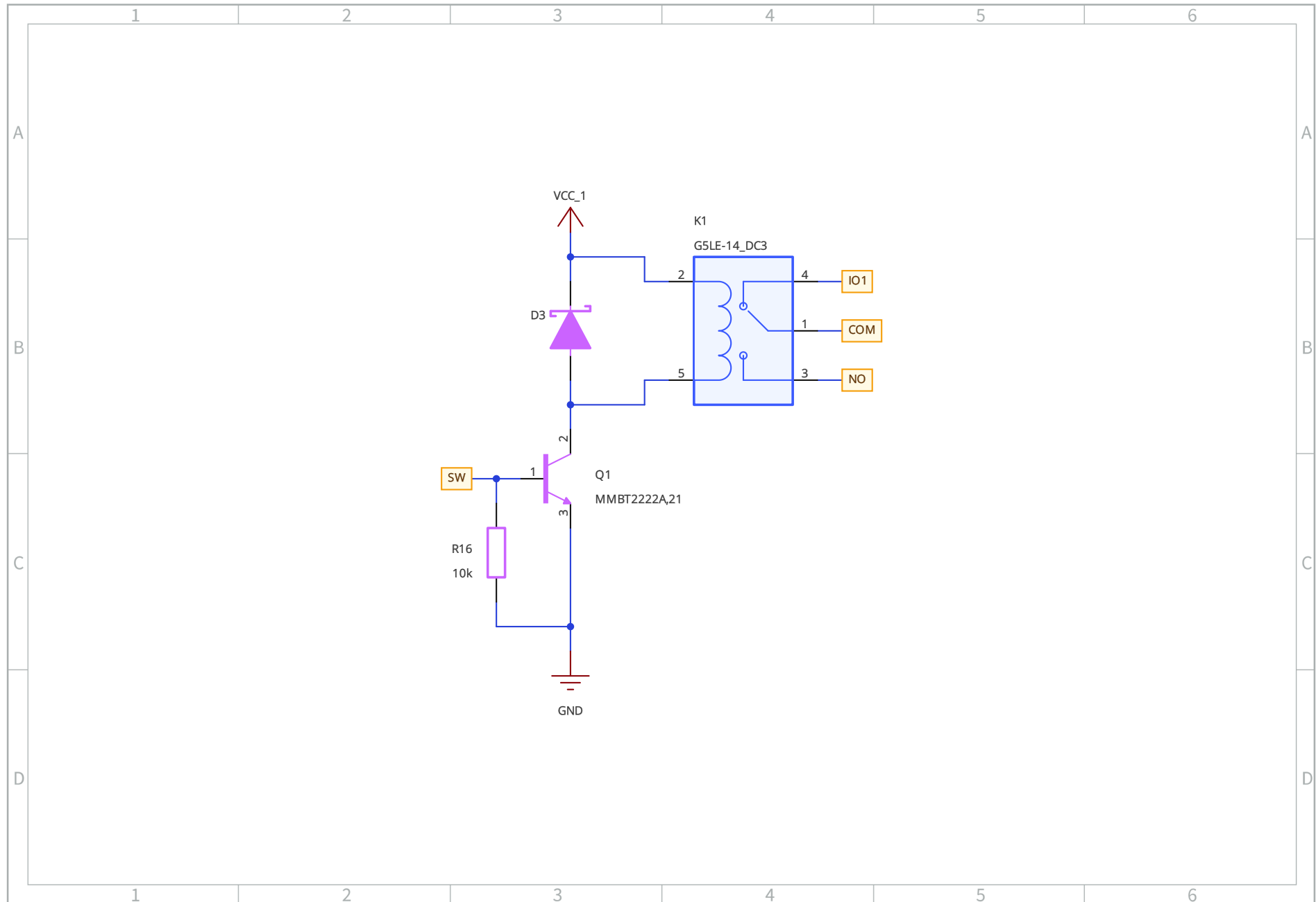


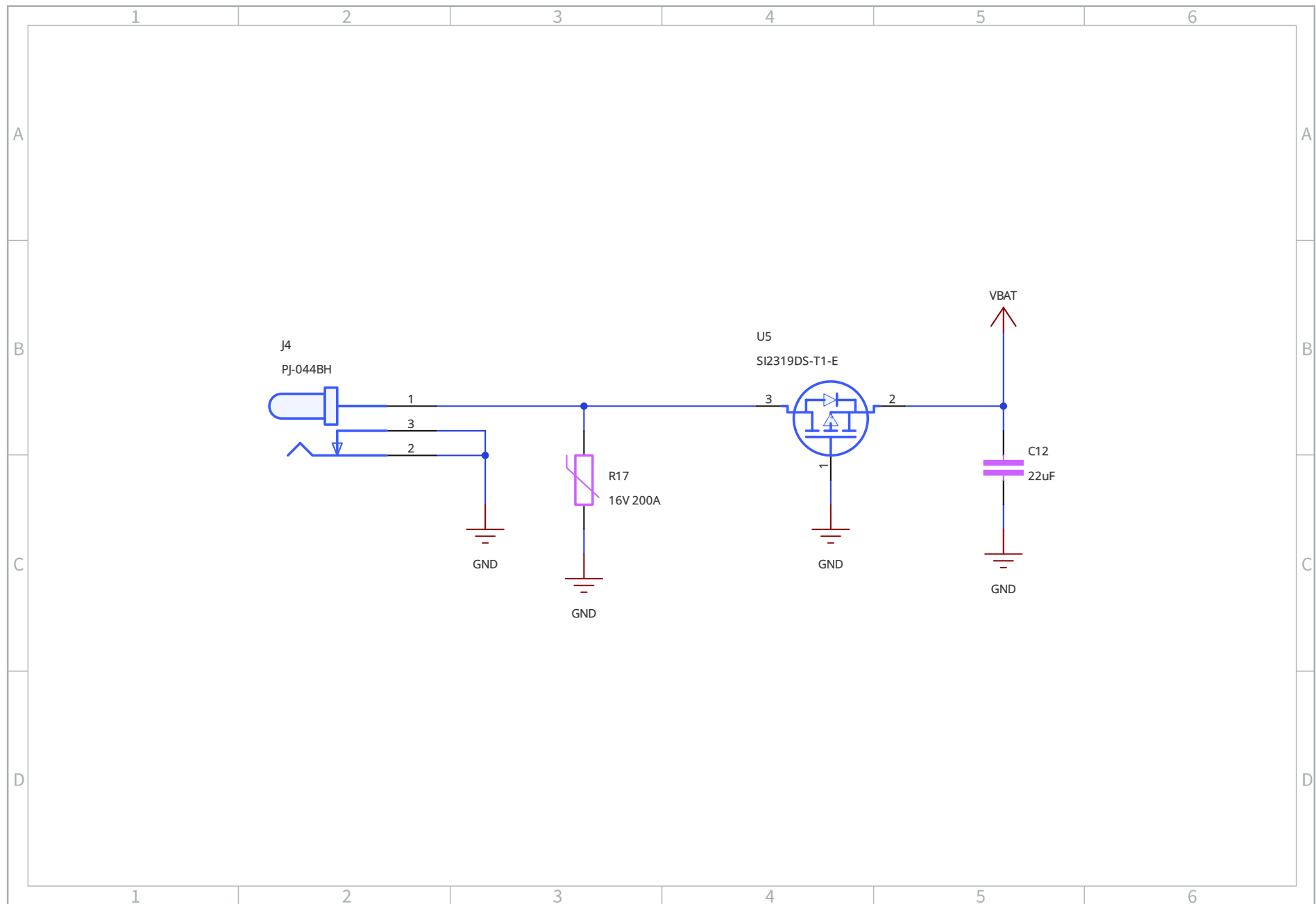


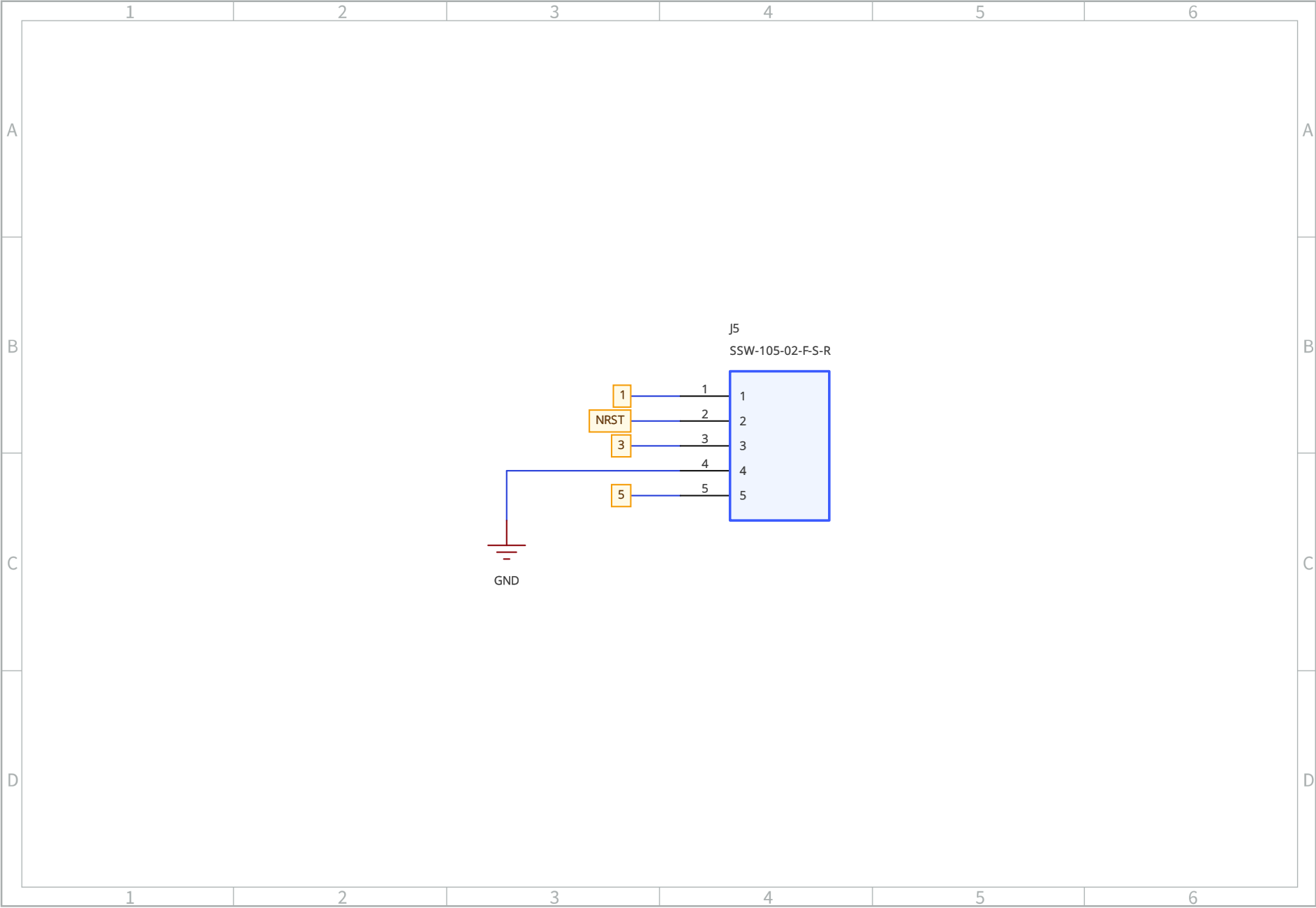


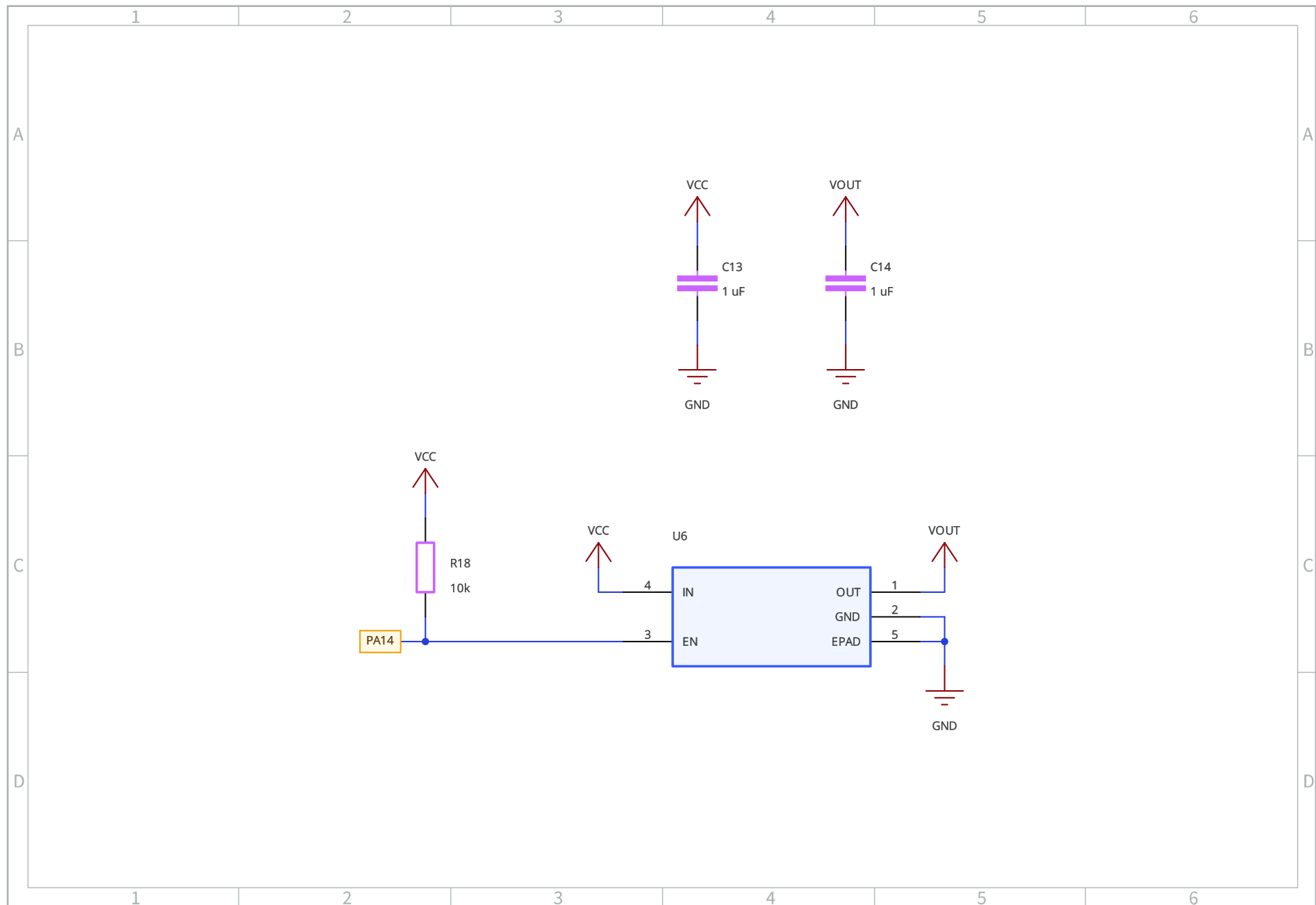


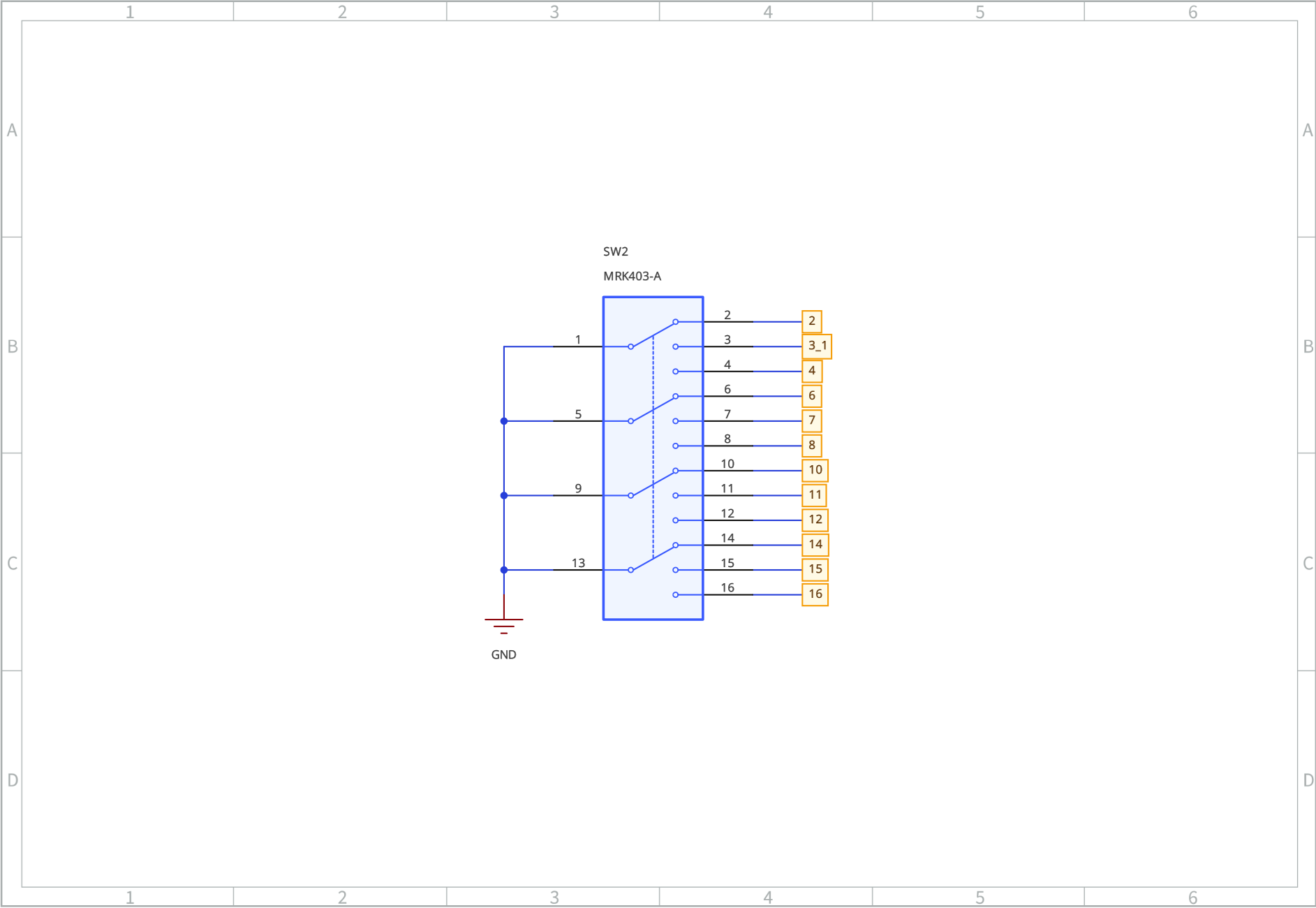












Ref. Des.	Value	Manufacturer	MPN	Package	Description
C1	100 nF	KYOCERA AVX COMPONENTS CORP	0402ZD104KAT2A	0402	Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 0.1uF, Surface Mount, 0402
C2	4.7 µF	MURATA MANUFACTURING CO LTD	GRM188R61E475KE11D	1608M/0603	Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 4.7uF, Surface Mount, 1608M/0603
C3	4.7 µF	MURATA MANUFACTURING CO LTD	GRM188R61E475KE11D	1608M/0603	Ceramic Capacitor, YES, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 4.7uF, Surface Mount, 1608M/0603
C4	100 nF	SAMSUNG ELECTRO-MECHANICS	CL05B104KO5NNNC	0402	Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402
C5	1 µF	SAMSUNG ELECTRO-MECHANICS	CL05A105KO5NNNC	0402	Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0402
C6	10 µF	SAMSUNG ELECTRO-MECHANICS	CL31A106KAHNNNE	1206	Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X5R, 15% TC, 10uF, Surface Mount, 1206
C7	1 µF	SAMSUNG ELECTRO-MECHANICS	CL05A105KO5NNNC	0402	Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0402
C8	1 µF	KYOCERA AVX COMPONENTS CORP	12061C105K4Z4A	1206	Ceramic Capacitor, Multilayer, Ceramic, 100V, 10% +Tol, 10% -Tol, X7R, 15% TC, 1uF, Surface Mount, 1206
C9	100 nF	KYOCERA AVX COMPONENTS CORP	08051C104KAT2A	0805	Ceramic Capacitor, Multilayer, Ceramic, 100V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0805
C10	0.47 µF	KYOCERA AVX COMPONENTS CORP	08051C474K4Z2A	0805	Ceramic Capacitor, Multilayer, Ceramic, 100V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.47uF, Surface Mount, 0805
C11	4.7 µF	KYOCERA CORP	CM05X5R475M25AH	0402	Ceramic Capacitor, Multilayer, Ceramic, 25V, 20% +Tol, 20% -Tol, X5R, 15% TC, 4.7uF, Surface Mount, 0402
C12	22 µF	MURATA MANUFACTURING CO LTD	GRT21BR61E226ME13L	2012M/0805	Ceramic Capacitor, Multilayer, Ceramic, 25V, 20% +Tol, 20% -Tol, X5R, 15% TC, 22uF, Surface Mount, 2012M/0805
C13	1 µF	SAMSUNG ELECTRO-MECHANICS	CL05A105KP5NNWC	0402	Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0402
C14	1 µF	SAMSUNG ELECTRO-MECHANICS	CL05A105KP5NNWC	0402	Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0402
D1		ONSEMI	FSV1045V		Rectifier Diode, Schottky, 1 Phase, 1 Element, 10A, 45V V(RRM), Silicon, TO-277

Ref. Des.	Value	Manufacturer	MPN	Package	Description
D2		STMICROELECTRONICS	STPS3H100U		Rectifier Diode, Schottky, 1 Phase, 1 Element, 3A, 100V V(RRM), Silicon
D3		ONSEMI	MBR140SFT1G		Rectifier Diode, Schottky, 1 Phase, 1 Element, 1A, 40V V(RRM), Silicon
J1		WUERTH ELEKTRONIK GMBH & CO KG	691214110002S		Barrier Strip Terminal Block
J2	MALE	HIROSE ELECTRIC CO LTD	U.FL-R-SMT-1(01)		RF Connector, 1 Contact(s), Male, Board Mount, Surface Mount Terminal, Receptacle
J3		WUERTH ELEKTRONIK GMBH & CO KG	691214110002S		Barrier Strip Terminal Block
J4		CUI DEVICES	PJ-044BH		Circular DC Power Connector, 24V, 5A, 2.5mm Dia
J5	FEMALE	SAMTEC INC	SSW-105-02-F-S-RA		Board Connector, 5 Contact(s), 1 Row(s), Female, Right Angle, 0.1 inch Pitch, Solder Terminal, Locking, Socket
K1		OMRON CORP	G5LE-14-DC3		Power/Signal Relay, SPDT, Momentary, 0.136A (Coil), 3VDC (Coil), 400mW (Coil), 8A (Contact), 125VDC (Contact), DC Input, AC/DC Output
Q1		NEXPERIA	MMBT2222A,215		Small Signal Bipolar Transistor, 0.6A I(C), 40V V(BR)CEO, 1-Element, NPN, Silicon, TO-236AB
R1	47 kOhm	VISHAY INTERTECHNOLOGY INC	CRCW060347K0FKEA	0603	Fixed Resistor, Metal Glaze/Thick Film, 0.1W, 47000ohm, 75V, 1% +/-Tol, Surface Mount, 0603
R2	47 kOhm	VISHAY INTERTECHNOLOGY INC	CRCW060347K0FKEA	0603	Fixed Resistor, Metal Glaze/Thick Film, 0.1W, 47000ohm, 75V, 1% +/-Tol, Surface Mount, 0603
R3	47 kOhm	VISHAY INTERTECHNOLOGY INC	CRCW060347K0FKEA	0603	Fixed Resistor, Metal Glaze/Thick Film, 0.1W, 47000ohm, 75V, 1% +/-Tol, Surface Mount, 0603
R4	47 kOhm	VISHAY INTERTECHNOLOGY INC	CRCW060347K0FKEA	0603	Fixed Resistor, Metal Glaze/Thick Film, 0.1W, 47000ohm, 75V, 1% +/-Tol, Surface Mount, 0603
R5	470 Ohm	YAGEO CORP	RC0402FR-07470RL	0402	Fixed Resistor, Metal Glaze/thick Film, 0.0625W, 470ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402
R6	10 kOhm	SAMSUNG ELECTRO-MECHANICS	RC1005F103CS	0402	Fixed Resistor, Metal Glaze/thick Film, 0.0625W, 10000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402

Ref. Des.	Value	Manufacturer	MPN	Package	Description
R7	10 kOhm	SAMSUNG ELECTRO-MECHANICS	RC1005F103CS	0402	Fixed Resistor, Metal Glaze/thick Film, 0.0625W, 10000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402
R8	100 mOhm	PANASONIC CORP	ERJ2BSFR10X	0402	Fixed Resistor, Metal Glaze/thick Film, 0.166W, 0.1ohm, 0.1288V, 1% +/-Tol, 0,300ppm/Cel, Surface Mount, 0402
R9	2 kOhm	PANASONIC CORP	ERJ2RKF2001X	0402	Fixed Resistor, Metal Glaze/thick Film, 0.1W, 2000ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402
R10	22 Ohm	PANASONIC CORP	ERJ2RKF22R0X	0402	Fixed Resistor, Metal Glaze/thick Film, 0.1W, 22ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402
R11	22 Ohm	PANASONIC CORP	ERJ2RKF22R0X	0402	Fixed Resistor, Metal Glaze/thick Film, 0.1W, 22ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402
R12	22 Ohm	PANASONIC CORP	ERJ2RKF22R0X	0402	Fixed Resistor, Metal Glaze/thick Film, 0.1W, 22ohm, 50V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0402
R13	470 Ohm	YAGEO CORP	AC0402FR-7W470RL	0402	Fixed Resistor, Metal Glaze/Thick Film, 0.125W, 470ohm, 75V, 1% +/-Tol, Surface Mount, 0402
R14	10 kOhm	KOA SPEER ELECTRONICS INC	SG73P1EWTPP1002F	0402	Fixed Resistor, Metal Glaze/Thick Film, 0.25W, 10000ohm, 75V, 1% +/-Tol, Surface Mount, 0402
R15	10 kOhm	KOA SPEER ELECTRONICS INC	SG73P1EWTPP1002F	0402	Fixed Resistor, Metal Glaze/Thick Film, 0.25W, 10000ohm, 75V, 1% +/-Tol, Surface Mount, 0402
R16	10 kOhm	PANASONIC CORP	ERJ2GEJ103X	0402	Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402
R17		SFI ELECTRONICS TECHNOLOGY INC	0805SA240-1R5J		SFI ELECTRONICS TECHNOLOGY INC - 0805SA240-1R5J
R18	10 kOhm	PANASONIC CORP	ERJ2GEJ103X	0402	Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402
SW1		OMRON CORP	B3U-1000P		Keypad Switch, 1 Switches, SPST, Momentary, 0.05A, 12VDC, 1.99N, Solder Terminal, Surface Mount-straight
SW2		NKK SWITCHES CO LTD	MRK403-A		Rotary Switch
U1		STMICROELECTRONICS	VL53L5CXV0GC/1		VCSEL TOF Photoelectric Sensor, 20mm Min, 4000mm Max, 0.4-2.1V, Rectangular, Surface Mount

Ref. Des.	Value	Manufacturer	MPN	Package	Description
U2		TEXAS INSTRUMENTS INC	BQ24210DQCT		Adjustable, 1 Channel, +3.3V, PDSO10
U3		ANALOG DEVICES INC	LTC2959IDDB#TRMPBF		Adjustable, 1 Channel, +1.45V, CMOS
U4		Seeed Studio	317990687		Seeed Studio - 317990687
U5		VISHAY INTERTECHNOLOGY INC	SI2319DS-T1-E3		Power Field-Effect Transistor, 2.3A I(D), 40V, 0.082ohm, 1-Element, P-Channel, Silicon, Metal-oxide Semiconductor FET, TO-236AB
U6		ONSEMI	NCP114AMX330TBG		Fixed Positive LDO Regulator, 3.3V, 0.2V Dropout, CMOS, PDSO4